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import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_excel("/content/Housing.xlsx")

print(df.head())
print(df.info())
print(df.describe())

print(df.isnull().sum())

fig, axs = plt.subplots(2, 3, figsize=(10, 5))

sns.boxplot(x=df['price'], ax=axs[0, 0])
sns.boxplot(x=df['area'], ax=axs[0, 1])
sns.boxplot(x=df['bedrooms'], ax=axs[0, 2])
sns.boxplot(x=df['bathrooms'], ax=axs[1, 0])
sns.boxplot(x=df['stories'], ax=axs[1, 1])
sns.boxplot(x=df['parking'], ax=axs[1, 2])

plt.tight_layout()
plt.show()

sns.pairplot(df)
plt.show()

df_encoded = df.copy()

categorical_cols = ['mainroad', 'guestroom', 'basement',
                    'hotwaterheating', 'airconditioning',
                    'prefarea', 'furnishingstatus']

df_encoded = pd.get_dummies(df_encoded, columns=categorical_cols,
                             drop_first=True)

X = df_encoded.drop('price', axis=1)
y = df_encoded['price']

X_train, X_test, y_train, y_test = train_test_split(

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    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R-squared:", r2)

plt.figure(figsize=(10, 6))
plt.scatter(y_test, y_pred, alpha=0.5)
plt.plot([y.min(), y.max()], [y.min(), y.max()], 'r--')
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted Prices")
plt.show()

plt.figure(figsize=(10, 6))
sns.residplot(x=y_pred, y=y_test - y_pred, lowess=True)
plt.axhline(0, linestyle='--', color='red')
plt.xlabel("Predicted Values")
plt.ylabel("Residuals")
plt.title("Residual Plot")
plt.show()

```

	price	area	bedrooms	bathrooms	stories	mainroad	guestroom
basement \							
0	13300000	7420	4	2	3	yes	no
no							
1	12250000	8960	4	4	4	yes	no
no							
2	12250000	9960	3	2	2	yes	no
yes							
3	12215000	7500	4	2	2	yes	no
yes							
4	11410000	7420	4	1	2	yes	yes
yes							

	hotwaterheating	airconditioning	parking	prefarea	furnishingstatus
0	no	yes	2	yes	furnished
1	no	yes	3	no	furnished
2	no	no	2	yes	semi-furnished

3	no	yes	3	yes	furnished
4	no	yes	2	no	furnished

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 545 entries, 0 to 544

Data columns (total 13 columns):

#	Column	Non-Null Count	Dtype
0	price	545 non-null	int64
1	area	545 non-null	int64
2	bedrooms	545 non-null	int64
3	bathrooms	545 non-null	int64
4	stories	545 non-null	int64
5	mainroad	545 non-null	object
6	guestroom	545 non-null	object
7	basement	545 non-null	object
8	hotwaterheating	545 non-null	object
9	airconditioning	545 non-null	object
10	parking	545 non-null	int64
11	prefarea	545 non-null	object
12	furnishingstatus	545 non-null	object

dtypes: int64(6), object(7)

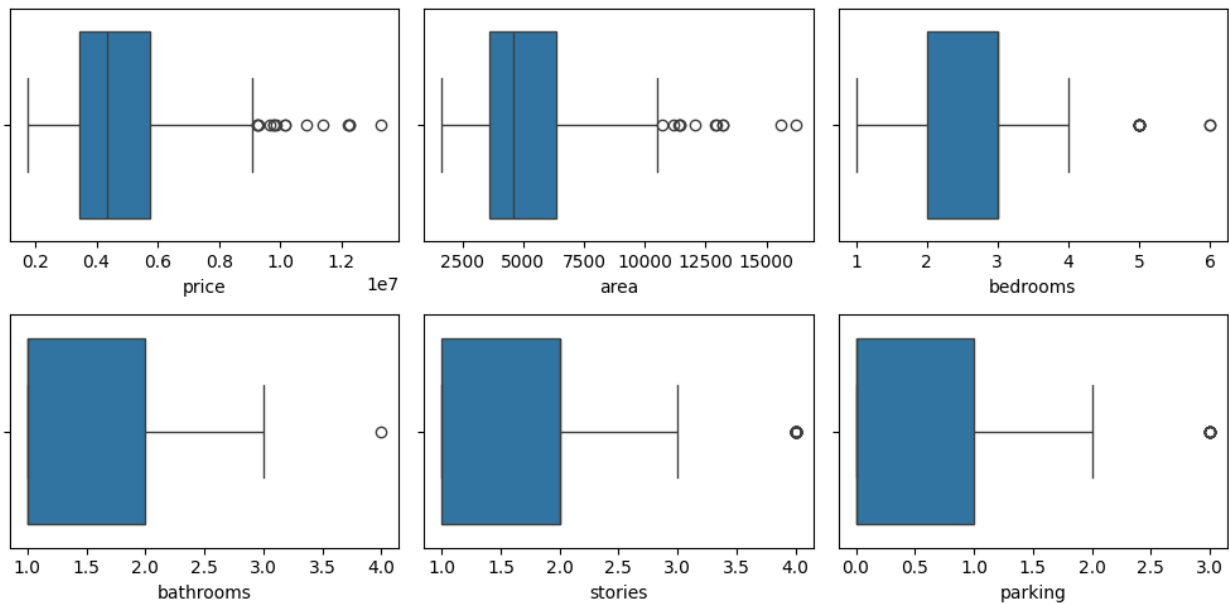
memory usage: 55.5+ KB

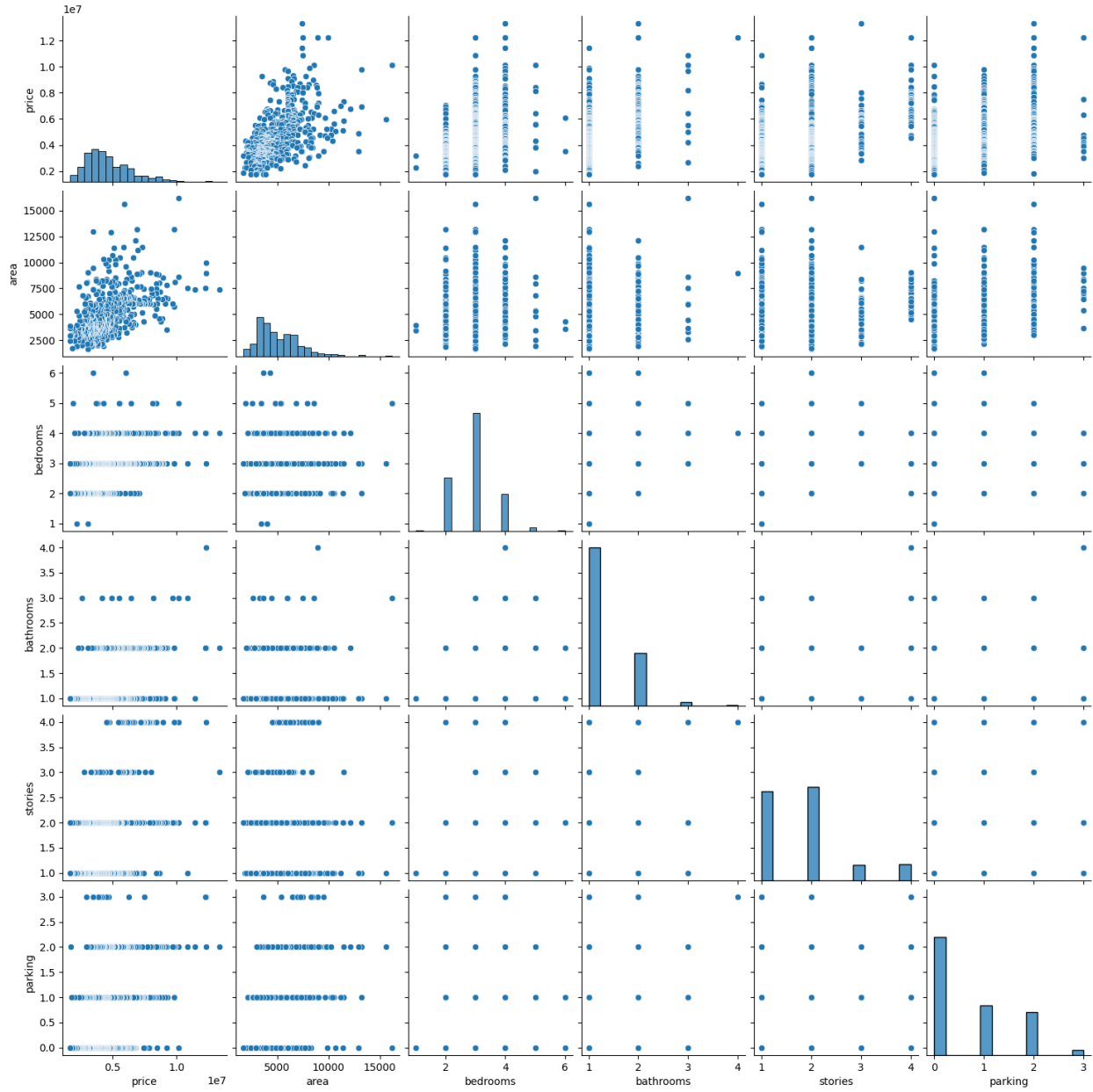
None

	price	area	bedrooms	bathrooms	stories
count	5.450000e+02	545.000000	545.000000	545.000000	545.000000
mean	4.766729e+06	5150.541284	2.965138	1.286239	1.805505
std	1.870440e+06	2170.141023	0.738064	0.502470	0.867492
min	1.750000e+06	1650.000000	1.000000	1.000000	1.000000
25%	3.430000e+06	3600.000000	2.000000	1.000000	1.000000
50%	4.340000e+06	4600.000000	3.000000	1.000000	2.000000
75%	5.740000e+06	6360.000000	3.000000	2.000000	2.000000
max	1.330000e+07	16200.000000	6.000000	4.000000	4.000000

	parking
count	545.000000
mean	0.693578
std	0.861586
min	0.000000
25%	0.000000
50%	0.000000
75%	1.000000

```
max      3.000000
price    0
area     0
bedrooms 0
bathrooms 0
stories  0
mainroad 0
guestroom 0
basement 0
hotwaterheating 0
airconditioning 0
parking  0
prefarea 0
furnishingstatus 0
dtype: int64
```





Mean Squared Error: 1754318687330.6638
R-squared: 0.6529242642153184

